Semiconductor Properties

1.3 Electron and hole densities in equilibrium

1.3.1 Distribution of quantum states in the energy band

To develop quantitative expressions for electron and hole densities (n and p) we shall introduce two mathematical results without proof. The first of these results is from quantum mechanics and it deals with the distribution of quantum states in energy in the conduction band, the energy gap and the valence band of the semiconductor. According to this, the number of states (a state is an energy level which an electron can occupy) in the conduction band with energies between E and E+dE is given by

$$g(E)dE = 4\pi(2\, m_e^*/h^2)^{3/2}(E-E_C)^{1/2}\, dE \text{ for } E > E_C \quad \ldots\ldots.(1.10a)$$

Similarly, the number of states in the valence band with energies between E and E+dE is given by

$$g(E)dE = 4\pi(2\, m_h^*/h^2)^{3/2}(E_V - E)^{1/2}\, dE \text{ for } E < E_V \ldots\ldots.(1.10b)$$

As there are no allowable states in the forbidden gap, $g(E) = 0$ for $E_V < E < E_C$. It is clear from the above equations that g(E) is the density of states, i.e. the number of states per unit volume per unit energy around an energy level E. Also, in the above equations, h is the Planck constant and **m_e^* and m_h^* are the effective masses of electrons and holes** respectively. Since the electrons in a crystal are not completely free, but instead interact with the periodic potential of the lattice, their wave-particle motion cannot be expected to be the same as for electrons in free space. Thus in applying usual equations of electrodynamics to charge carriers in a solid, we must use altered values of particle mass (the so-called “effective mass”). In doing so, we account for most of the influence of

lattice so that electrons and holes can be considered as 'almost free charge carriers' in most computations. The effective masses of electrons and holes are different in different materials. For example, $m_e^* = 0.067m_e$ in GaAs, where m_e is the free electron mass. From eqn. (1.10), we see that at the band edges, i.e. at $E = E_C$ and $E = E_V$, $g(E) = 0$. The density of states increases continuously as we move deeper into the bands, i.e. for increasing E in the conduction band and decreasing E in the valence band.

1.3.2 Fermi-Dirac statistics

The other result that we shall assume is the Fermi-Dirac distribution function. Electrons in solids follow Fermi-Dirac statistics, which can be expressed simply as follows: the probability f(E) that an electron occupies an available state at an energy level (E) at thermal equilibrium at a temperature T(K) is

$$f(E) = \frac{1}{1+\exp\left(\frac{E-E_F}{kT}\right)} \quad \text{.....(1.11)}$$

where k is Boltzmann's constant and E_F is the Fermi energy level. From eqn.(1.11), we see that the maximum value of f(E) is 1 when $E \rightarrow -\infty$ and the minimum value is 0 when $E \rightarrow \infty$. For an energy $E = E_F$, the occupation probability at any temperature is $f(E) = 0.5$. It may also be noted that since f(E) is the probability that a particular state is occupied, the probability that it is vacant is [1-f(E)].

Some properties of Fermi-Dirac Function are now listed as follows:

1. At T = 0K, the distribution function takes a simple rectangular form as shown in Fig.1.11. This implies that all states above E_F are empty and all states below E_F are filled with electrons, i.e. for $E > E_F$, $f(E) = 0$ and for $E < E_F$, $f(E) = 1$.

2. As shown in Fig.1.11, for T > 0K, some probability exists for states above E_F to be filled, i.e. f(E) is nonzero for $E > E_F$. There is also a corresponding probability that states below E_F are empty, since f(E) < 1 for $E < E_F$. For $E > E_F$, the state at a higher energy has a lesser probability of being occupied by an electron. On the other hand, for $E < E_F$, the state at a lower energy has a lesser probability of being vacant. Also, the probability that a state at $E > E_F$ is occupied or a state at $E < E_F$ is vacant increases at higher temperature.
3. It can be shown that the probability of a state at an energy ΔE above E_F being occupied is exactly the same as the probability that a state at an energy ΔE below E_F is vacant. In other words, f(E) is symmetrical about E_F for all temperatures, i.e. $f(E_F + \Delta E) = 1 - f(E_F - \Delta E)$.
4. A filled state in the conduction band indicates the presence of a free electron and a vacant state in the valence band indicates the presence of a hole. The free electron density around an energy level E in the conduction band would be the product of the density of states g(E) and the probability f(E) that the states are occupied. On the other hand, the density of holes around an energy level E in the valence band would be the product of the density of states g(E) and the probability [1 - f(E)] that the states are vacant. Since the number of free electrons is the same as the number of holes in an intrinsic semiconductor, the symmetry of f(E) about E_F implies that for an intrinsic semiconductor, if the density of available states in the conduction and valence band are the same, then the Fermi level should lie exactly at the middle of the bandgap. This is shown clearly in Fig. 1.12(a).

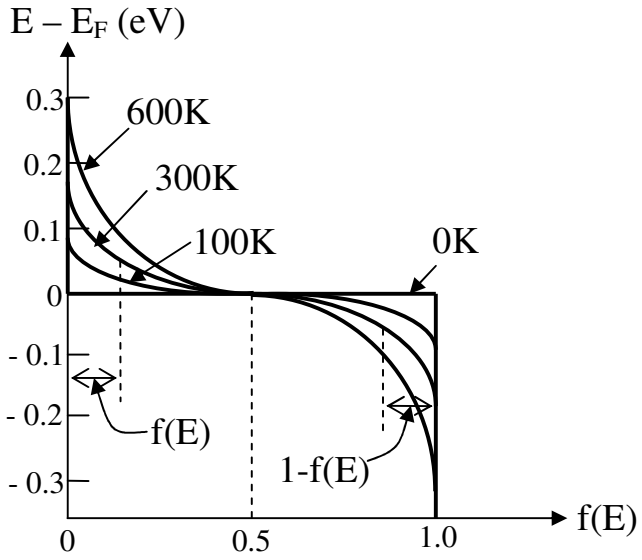


Fig. 1.11 The Fermi distribution function f(E) versus ($E - E_F$) at various temperatures

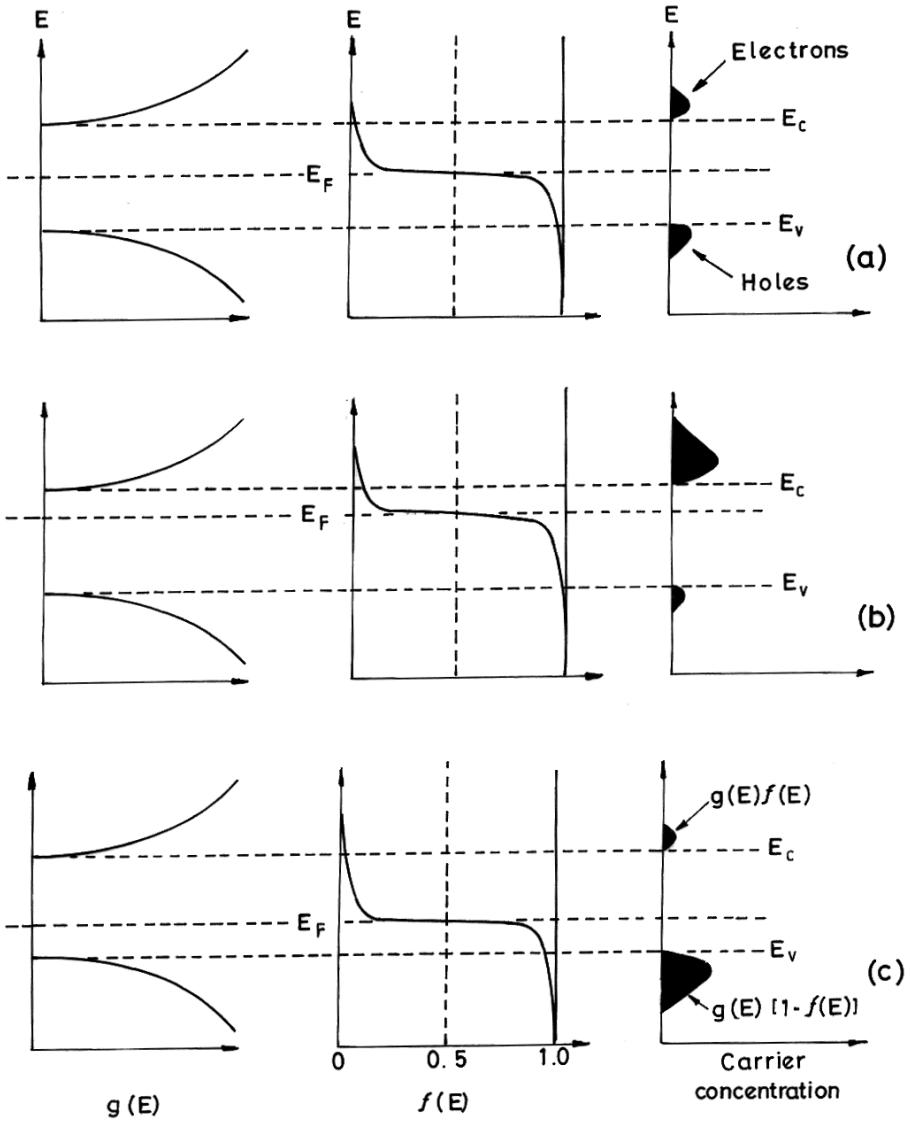


Fig. 1.12 The density of states g(E), the Fermi distribution function f(E) and the electron and hole concentrations in (a) intrinsic semiconductor, (b) n-type semiconductor and (c) p-type semiconductor

5. In an n-type material, n > p. This implies that there are more filled states in the conduction band than there are empty states in the valence band. From the symmetry of f(E), this means that E_F should be closer to E_C for n-type material. As shown in Fig. 1.12(b), in this case, the area under the g(E)f(E) curve in the conduction band

(which represents the concentration of electrons in the conduction band) is more than the g(E)[1-f(E)] curve in the valence band (which represents the concentration of holes in the valence band). Using a similar argument, we can see from Fig. 1.12(c) that in a p-type material, where p > n, E_F should be closer to E_V than to E_C. For mathematical proof and further discussion of the last two points, refer to Section 1.3.5 and 1.3.6.

1.3.3 Electron Concentration in the conduction Band

Based on the above discussion, we can now write the electron concentration in the conduction band as

$$n = \int_{E_C}^{E_{top}} f(E)g(E)dE \quad \text{.....(1.12)}$$

where E_{top} is the energy at the top of the conduction band. Substituting eqns.(1.10a) and (1.11) into eqn.(1.12), we have

$$n = \int_{E_C}^{E_{top}} \frac{4\pi\left(2m_e^*/h^2\right)^{3/2}(E-E_C)^{1/2}}{1+\exp\left(\frac{E-E_F}{kT}\right)}dE \quad \text{.....(1.13)}$$

However, this equation is very difficult to solve analytically. Hence the following approximations are made:

i) Let us assume that $(E_C - E_F) >> 3kT$. Since kT is 0.026eV at room temperature, it is usually a good approximation for normal doping levels. Using this, since $(E - E_F) >> 3kT$ for any energy level E in the conduction band, f(E) given by eqn. (1.11) can be approximated as $f(E) \approx \exp[-(E-E_F)/kT]$. In other words, the Fermi-Dirac distribution can now be approximated by a Boltzmann distribution. However, it must

be remembered that when the semiconductor is very heavily doped (degenerate), E_F is very close to E_C and this approximation may not be valid.

ii) Typically, the width of the conduction band is several eV. However, for $E > E_C$, the probability of electron occupancy approaches zero when $(E - E_C)$ is of the order of kT. Hence, most of the electrons in the conduction band have energies close to E_C. So the upper limit of the integration in eqn. (1.13) can be changed to ∞ without any loss of accuracy.

Therefore eqn. (1.13) can now be modified as

$$n = \int_{E_C}^{\infty} 4\pi\left(2m_e^*/h^2\right)^{3/2}(E-E_C)^{1/2}\exp\left[-\left(\frac{E-E_F}{kT}\right)\right]dE$$
$$= \frac{4\pi}{h^3}\left(2m_e^*\right)^{3/2}\exp\left[-\left(\frac{E_C-E_F}{kT}\right)\right]\int_{E_C}^{\infty}(E-E_C)^{1/2}\exp\left[-\left(\frac{E-E_C}{kT}\right)\right]dE \quad \text{.....(1.14)}$$

The solution of eqn. (1.14) is given by

$$n = N_C\exp\left[-\left(\frac{E_C-E_F}{kT}\right)\right] \text{.....(1.15)}$$

where $N_C = 2(2\pi kTm_e^*/h^2)^{3/2}$ is called the **effective density of states in the conduction band.** The concept of N_C is as follows: from eqn. (1.15), we see that the expression of n is of the form $n = N_C f(E_C)$. Therefore, if all the states in the conduction band were to be confined at the particular energy level $E = E_C$, then N_C represents the number of states required to be present at E_C, so that after multiplying with the probability of electron occupancy at $E = E_C$ we get the number of electrons in the conduction band. However, it must be noted that the actual density of states at $E = E_C$ is zero and the concept of effective density of states only helps us to obtain a simple expression for the free electron

concentration in terms of the separation between the Fermi level and the conduction band edge. Since N_C depends on m_e^*, its value is different for different materials.

1.3.4 Hole concentration in the valence band

To compute the hole concentration in the valence band, we note that the probability of a hole occupying a level in the valence band (f_h) is actually the probability that a state is not occupied, or. $f_h(E) = 1-f(E)$. Hence, from eqn. (1.11), we can write that

$$f_h(E) = 1 - f(E) = \frac{\exp\left(\frac{E - E_F}{kT}\right)}{1 + \exp\left(\frac{E - E_F}{kT}\right)} = \frac{1}{1 + \exp\left(\frac{E_F - E}{kT}\right)} \quad \text{.....(1.16)}$$

The hole concentration in the valence band can now be written as

$$p = \int_{E_{bot}}^{E_V} f_h(E) g(E) dE \quad \text{.....(1.17)}$$

where E_{bot} is the energy corresponding to the bottom of the valence band. After making analogous assumptions as in the previous case, i.e. $(E_F\text{-}E_V) >> 3kT/q$ from which we have $f_h(E) \approx \exp[-(E_F\text{-}E)/kT]$ in the valence band and replacing E_{bot} with $-\infty$, we have

$$\begin{aligned} p &= \int_{-\infty}^{E_V} 4\pi \left(2m_h^*/h^2\right)^{3/2} (E_V - E)^{1/2} \exp\left[-\left(\frac{E_F - E}{kT}\right)\right] dE \\ &= \frac{4\pi}{h^3} \left(2m_h^*\right)^{3/2} \exp\left[-\left(\frac{E_F - E_V}{kT}\right)\right] \int_{-\infty}^{E_V} (E_V - E)^{1/2} \exp\left[-\left(\frac{E_V - E}{kT}\right)\right] dE \end{aligned} \quad \text{.....(1.18)}$$

From eqn.(1.18) we obtain the concentration of holes in valence band at thermal equilibrium as

$$p = N_V \exp\left[-\left(\frac{E_F - E_V}{kT}\right)\right] \quad \text{.....(1.19)}$$

where $N_V = 2(2\pi kTm_h^*/h^2)^{3/2}$ is referred to as the **effective density of states in the valence band.** From eqn. (1.19), we see that the expression of p is of the form $p = N_V f_h(E_V)$. Therefore N_V represents the number of states required to be present at E_V, so that after multiplying with the probability of vacancy at $E = E_V$ we get the number of holes in the valence band. From the expressions of N_C and N_V it can be seen that they both increase with temperature. Also as m_e^* and m_h^* do not have the same value, N_C and N_V are not equal (although they are of the same order). For a particular semiconductor, at a given temperature N_C and N_V are constants. In case of silicon at room temperature (300K), $N_C = 2.8\text{x}10^{19}/\text{cm}^3$ and $N_V = 1.02\text{x}10^{19}/\text{cm}^3$.

1.3.5 Carrier concentration in Intrinsic Semiconductor

If E_i denotes the position of the Fermi energy level in an intrinsic semiconductor, then from eqns. (1.15) and eqn.(1.19) we have

$$n_i = N_C \exp\left[-\left(\frac{E_C - E_i}{kT}\right)\right] \text{.....(1.20)}$$

$$p_i = N_V \exp\left[-\left(\frac{E_i - E_V}{kT}\right)\right] \text{.....(1.21)}$$

where n_i and p_i denote the electron and hole concentrations for the intrinsic material. Since the electron concentration is equal to the hole concentration in an intrinsic semiconductor, $n_i = p_i$. Therefore from eqns. (1.20) and (1.21), we obtain

$$E_i = \frac{E_C + E_V}{2} - \frac{kT}{2}\ln\left(\frac{N_C}{N_V}\right) \quad \text{.....(1.22)}$$

This mathematically proves our observation in Section 1.3.2 that if the effective densities of states in the conduction and valence bands are equal, the Fermi level for an intrinsic material would be located exactly at the centre of the bandgap. However, since the

quantity $\frac{kT}{2}\ln\left(\frac{N_C}{N_V}\right)$ is rather small (a few meV) in practical situations, **E_i is assumed to be located approximately at the middle of the bandgap for all intrinsic materials.**

Again, since n_i is equal to p_i, from eqns. (1.20) and (1.21), we can write

$$n_i.p_i = n_i^2 = N_C N_V \exp\left[-\left(\frac{E_C - E_V}{kT}\right)\right] = N_C N_V \exp\left[-\left(\frac{E_g}{kT}\right)\right] \text{.....(1.23)}$$

Therefore

$$n_i = \sqrt{N_C N_V}\exp\left[-\left(\frac{E_g}{2kT}\right)\right] \text{.....(1.24)}$$

So, we see that the intrinsic carrier concentration for a particular semiconductor is constant at a given temperature. **The value of the intrinsic carrier concentration is a strong function of the bandgap and is higher for lower bandgap materials.** This agrees very well with our discussion in Section 1.1. Since the bandgap is actually the energy required to break a covalent bond, a smaller bandgap would imply a larger number of broken bonds at room temperature and consequently a larger number of carriers. The bandgap and the intrinsic carrier concentration for a few semiconductors at room temperature are given in Table 1.3.

Semiconductor	**Bandgap (eV)**	**Intrinsic carrier concentration (/cm^3)**
Indium Arsenide	0.35	1.3×10^{15}
Germanium	0.66	2.3×10^{13}
Silicon	1.12	1.5×10^{10}
Indium Phosphide	1.34	1.2×10^{8}
Gallium Arsenide	1.42	1.8×10^{6}

Table 1.3 Bandgap and intrinsic carrier concentrations in some common semiconductors

Q 1.2 How does the intrinsic carrier concentration vary with temperature?

In the expression for n_i given by eqn.(1.24), in addition to the presence of temperature (T) as a variable, N_C, N_V and E_g are all functions of temperature. Ignoring the small variation of bandgap with temperature, but considering the expressions for N_C and N_V together with eqn.(1.24), we can write

$$n_i \propto T^{3/2} \exp\left[-\left(\frac{E_g}{2kT}\right)\right]$$

Thus we see that n_i increases very rapidly with temperature, the increase being almost exponential in nature. This again agrees with our discussion in Section 1.1. With increase in temperature, more bonds are broken giving rise to more free carriers. Fig.1.13 shows the variation of n_i with temperature for Ge, Si and GaAs.

Example 1.1 The intrinsic carrier concentration of silicon (E_g = 1.12eV) at 300K is 1.5 x $10^{10}/cm^3$. Calculate n_i for Si at 400K and 500K. (k = 8.62 x 10^{-5} eV/K)

Since $n_i \propto T^{3/2} \exp\left[-\left(\frac{E_g}{2kT}\right)\right]$, we can write

$$\frac{(n_i)_{400K}}{(n_i)_{300K}} = \left(\frac{400}{300}\right)^{3/2} \exp\left[-\left\{\frac{1.12}{2x8.62x10^{-5}}\left(\frac{1}{400}-\frac{1}{300}\right)\right\}\right] = 1.54 \text{ x } 224.475 = 345.7$$

Therefore $(n_i)_{400K} = 1.5 \text{ x } 10^{10} \text{ x } 345.7 = 5.185 \text{ x } 10^{12} /cm^3$.

$$\text{Similarly } (n_i)_{500K} = 1.5x10^{10} x\left(\frac{500}{300}\right)^{3/2} \exp\left[-\left\{\frac{1.12}{2x8.62x10^{-5}}\left(\frac{1}{500}-\frac{1}{300}\right)\right\}\right]$$

$$= 1.5x\ 10^{10} \text{ x } 2.15 \text{ x } 5779.23 = 1.86 \text{ x } 10^{14} /cm^3.$$

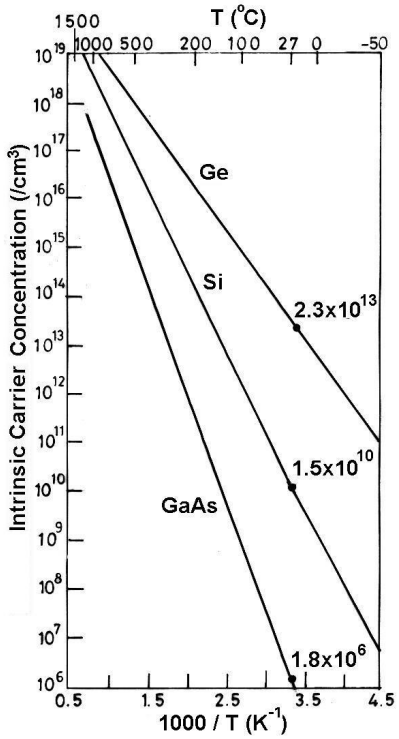


Fig.1.13 Intrinsic carrier concentrations in Ge, Si and GaAs as a function of temperature

1.3.6 Position of Fermi level in Extrinsic Semiconductors

From eqns. (1.15) and (1.19), we get

$$n.p = N_C N_V \exp\left[-\left(\frac{E_C - E_V}{kT}\right)\right] = N_C N_V \exp\left[-\left(\frac{E_g}{kT}\right)\right] \ldots\ldots\ldots\ldots (1.25)$$

From eqns. (1.24) and (1.25) we now have

$$n.p = n_i^2 \ldots\ldots\ldots (1.26)$$

This is an important fundamental relationship, which states that **for any semiconductor at thermal equilibrium, the product of electron and hole concentrations is a constant** and is independent of doping. From eqns. (1.15) and (1.20), it can also be written that

$$n = n_i \exp\left(\frac{E_F - E_i}{kT}\right) \ldots\ldots\ldots (1.27)$$

and similarly from eqns. (1.19) and (1.21), we have

$$p = n_i \exp\left(\frac{E_i - E_F}{kT}\right) \text{.....(1.28)}$$

From the above expressions, the position of the Fermi level in an extrinsic semiconductor can be easily obtained. Rearranging eqns. (1.27) and (1.28), we can also write

$$E_F = E_i + kT \ln\left(\frac{n}{n_i}\right) \text{.....(1.29)}$$

$$E_F = E_i - kT \ln\left(\frac{p}{n_i}\right) \text{.....(1.30)}$$

As already discussed, for an n-type (p-type) semiconductor, the electron (hole) concentration is much larger than the intrinsic carrier concentration. Therefore, eqns. (1.29) and (1.30) mathematically prove that the Fermi energy level for an n-type semiconductor is above the intrinsic Fermi level (closer to the conduction band edge) while for a p-type semiconductor, the Fermi level is below the intrinsic level (closer to the valence band edge), as already pointed out in Section 1.3.2.

Q 1.3 What is the physical reason for the relation $n.p = n_i^2$?

For any equilibrium condition, the rate of generation of carriers must be equal to the recombination rate. The generation rate in a particular semiconductor at a given temperature is a constant, irrespective of whether the semiconductor is intrinsic or extrinsic. This implies that the recombination rate at a particular temperature must also be a constant for the semiconductor, whether intrinsic or extrinsic. Since the recombination rate depends on the product of the electron and hole concentration, we can infer that the product of the electron and hole concentration in an extrinsic semiconductor must be equal to that in the intrinsic case. Hence $n.p = n_i^2$.

Example 1.2: In an n-type silicon sample, the Fermi level is 0.3 eV below the conduction band edge. Find the electron and hole concentration in the sample at room temperature (300K). (For Si, E_g = 1.1eV, $n_i = 1.5 \times 10^{10}/cm^3$ and $k=8.62 \times 10^{-5}$eV/K)

Since the intrinsic Fermi level is located approximately at the centre of the band gap, we can write $E_C - E_i = Eg/2 = 0.55$ eV for silicon. Also, it is given that $E_C - E_F = 0.3$eV.

Therefore, $E_F - E_i = (E_C - E_i) - (E_C - E_F) = 0.55 - 0.3 = 0.25$eV

From eqn. (1.27), we get

$$n = 1.5x10^{10} \exp\left(\frac{0.25}{8.62x10^{-5} x300}\right) = 1.5x10^{10} x1.58x10^{4} = 2.37x10^{14}/cm^3$$

From eqn.(1.26), $p = n_i^2/n = (1.5x10^{10})^2/(2.37x10^{14}) = 9.5x10^5/cm^3$

1.3.8 Equilibrium electron and hole concentration

For a homogeneous non-degenerate semiconductor with N_A acceptors and N_D donors per cm^3, from charge neutrality condition, equating the number of positive charges to the number of negative charges, we can write

$$N_D^+ + p = N_A^- + n \quad \ldots\ldots\ldots (1.39)$$

Assuming complete ionisation (justifiable when the temperature is close to 300K or above), this can be rewritten as

$$n - p = N_D^+ - N_A^- \approx N_D - N_A \quad \ldots\ldots\ldots (1.40)$$

Although we have assumed complete ionization in this Section, in the cases of incomplete ionization, N_D and N_A must be replaced by $\eta_D N_D$ and $\eta_A N_A$ in the following equations. From eqn. (1.26), we know that $np = n_i^2$. Therefore, substituting $p = n_i^2/n$ in eqn.(1.40), we have the quadratic equation

$$n^2 - (N_D - N_A)n - n_i^2 = 0 \quad \ldots\ldots\ldots (1.41)$$

The solution of eqn. (1.41) is

$$n = \frac{(N_D - N_A) + \sqrt{(N_D - N_A)^2 + 4n_i^2}}{2} \quad \ldots\ldots (1.42)$$

Similarly, replacing $n = n_i^2/p$ in eqn.(1.40) and solving for p, we have

$$p = \frac{(N_A - N_D) + \sqrt{(N_D - N_A)^2 + 4n_i^2}}{2} \quad \ldots\ldots (1.43)$$

Equations (1.42) and (1.43) are the general expressions for the electron and hole concentrations in a semiconductor. Let us now consider the specific cases of intrinsic, n-type and p-type semiconductors.

Case1: Intrinsic semiconductor- In this case as there are no donor or acceptor impurities, $N_D = N_A = 0$. So from equations (1.42) and (1.43), we see that $n = p = n_i$.

Case2: n-type semiconductor - For n-type material, $N_D >> N_A$. Assuming that $(N_D - N_A) >> 2n_i$, $n \approx N_D$ and $p \approx n_i^2/N_D$.

Case3: p-type semiconductor – For p-type material, $N_A >> N_D$. Assuming that $(N_A - N_D) >> 2n_i$, $p \approx N_A$ and $n \approx n_i^2/N_A$.

Materials that satisfy the condition $|N_D - N_A| >> 2n_i$ are called **strongly extrinsic material**. However, one interesting fact is that since n_i is a function of temperature, a material that is strongly extrinsic at one temperature may become nearly intrinsic as the temperature is increased. The following example will clarify this point.

Example1.5: Consider a silicon sample with electron concentration $n = 10^{13}/cm^3$ at 300K. Show that this material is strongly extrinsic at 300K, but becomes nearly intrinsic at 473K.

At 300K, n_i for silicon is $1.5x10^{10}/cm^3$. Therefore the hole concentration in this silicon sample is $p = n_i^2/n = 2.25x10^7/cm^3$. Assuming complete ionisation at this temperature and using eqn. (1.40), we get $N_D - N_A = n - p = 10^{13} - 2.25x10^7 \approx 10^{13}/cm^3$. This is the net dopant concentration and its value is independent of temperature. At 300K, evidently $(N_D-N_A) >> 2n_i$ and therefore the material is deemed to be strongly extrinsic.

At 473K, however, the value of n_i in silicon becomes $1.5x10^{14}/cm^3$. Obviously at this temperature N_D-N_A is much less than n_i and the material can no longer be deemed strongly extrinsic. From eqns. (1.42) and (1.43), the values of n and p at 473K are calculated to be $n = 1.55x10^{14}/cm^3$ and $p = 1.45x10^{14}/cm^3$. Since the values of n and p are nearly equal to each other, the material displays nearly intrinsic behaviour at this temperature.

A plot of electron concentration versus temperature for an n-type semiconductor sample is shown in Fig 1.14. As already discussed, at low temperatures the electron concentration is low since complete ionisation has not taken place. As the temperature increases, the degree of ionisation of the donors increases and the electron concentration increases consequently. Then the electron concentration remains nearly constant over a range of temperature signifying that the ionisation is complete. This is the range of temperature at which the material is strongly extrinsic. Finally at higher temperatures, n_i increases, increasing the electron as well as the hole concentration as thermal generation of electron and hole pairs dominates. When the thermally generated electron and hole concentrations become comparable to the dopant concentration, the material exhibits nearly intrinsic behaviour.

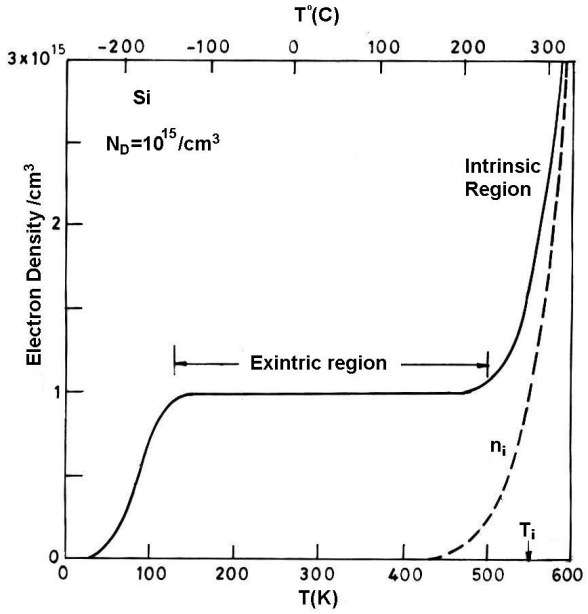


Fig. 1.14 Electron concentration as a function of temperature in an n-type silicon[1]. T_i is the intrinsic temperature [refer to Problem 6].

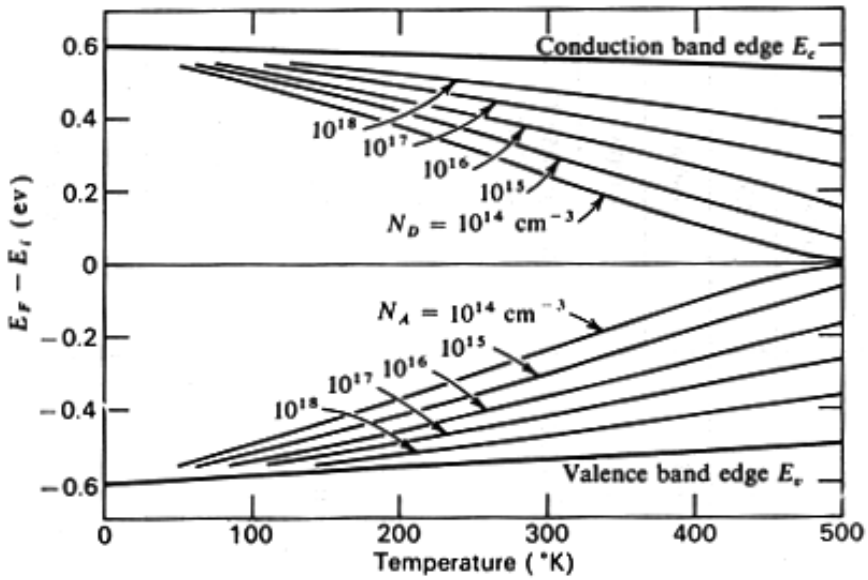


Fig.1.15 Fermi level positions in silicon as functions of temperature and impurity concentration[1] (From A.S. Grove, copyright © 1967 by John Wiley & Sons, Inc. Used by permission)

As a corollary, it will be interesting to note the variation in the position of the Fermi level with respect to the conduction band edge (i.e. $E_C - E_F$) with temperature. At very low temperatures, we know that the value of the ionisation coefficient is small, i.e. most of the donors are not ionised. In other words almost all donor states are "filled" or occupied by electrons. From our discussion of the Fermi energy in Section1.3.2, we can therefore reason that at such low temperature, Fermi level of this semiconductor must lie above the donor level, since levels below E_F are more likely to be occupied by electrons. Indeed it can be shown (given as Problem) that at very low temperature, for an n-type semiconductor, the position of the Fermi level is given by $E_F = \frac{E_C + E_D}{2} - \frac{kT}{2} \ln\left(\frac{2N_C}{N_D}\right)$.

As the temperature increases, the value of the ionisation coefficient approaches unity. This signifies that now the donor levels are "empty" and consequently Fermi level lies below the donor level. Then, at very high temperatures, when the electron and hole concentrations become nearly equal, the Fermi level comes close to the intrinsic Fermi level. Thus with increase in temperature, the position of the Fermi level shifts from a level close to the conduction band edge to approximately the middle of the bandgap. A similar line of reasoning can be adopted for a p-type material and it can be seen that for this case the position of the Fermi level varies from a level close to the valence band edge to the midgap (E_i) with increase in temperature. Fig.1.15 illustrates the nature of variation of Fermi level with temperature for various doping levels.

1.3.9 Concept of a constant Fermi level at thermal equilibrium

At this juncture, it may be useful to establish an important concept regarding the constancy of Fermi level at thermal equilibrium, i.e. when there is no net flow of charge.

This concept can be summarized as *"No discontinuity or gradient can exist in the Fermi Energy Level at thermal equilibrium"*. This concept will be particularly useful when we consider non-uniform doping, junctions between two semiconductors such as p-n junctions or metal-semiconductor junctions.

To prove the above concept, let us consider two dissimilar materials in intimate contact, so that electrons can flow between the two. Let $f_1(E)$ and $g_1(E)$ be the Fermi-Dirac distribution function and distribution of available states which can be occupied by electrons for material 1, while $f_2(E)$ and $g_2(E)$ are the Fermi-Dirac distribution function and distribution of available states for material 2. At thermal equilibrium, there is no current and therefore no net flow of charge. Also, there is no net transfer of energy. Therefore, for each energy E, transfer of electrons from material 1 to material 2 must be exactly balanced by a corresponding transfer of electrons from material 2 to material 1. The rate of transfer of electrons from material1 to material 2 at energy E will be proportional to the product of the number of filled states in material 1 and the number of empty states in material 2. Thus the electron flux from material 1 to material 2 is

$$F_{12} \propto g_1(E)f_1(E)\, g_2(E)\{1- f_2(E)\}\text{……..(1.44)}$$

where $f_1(E)$ and $f_2(E)$ are the Fermi-Dirac functions corresponding to material 1 and material 2 respectively. Similarly, the electron flux from material 2 to material 1 is

$$F_{21} \propto g_2(E)f_2(E)\, g_1(E)\{1- f_1(E)\}\text{……..(1.45)}$$

At equilibrium, $F_{12} = F_{21}$, i.e.

$$g_1(E)f_1(E)\, g_2(E)\{1- f_2(E)\} = g_2(E)f_2(E)\, g_1(E)\{1- f_1(E)\}\text{……(1.46)}$$

which leads to $f_1(E) = f_2(E)$, i.e.

$$\frac{1}{1+\exp\left(\frac{E-E_{F1}}{kT}\right)} = \frac{1}{1+\exp\left(\frac{E-E_{F2}}{kT}\right)} \ldots\ldots\ldots\ldots(1.47)$$

where E_{F1} and E_{F2} are Fermi energy levels in material 1 and material 2 respectively. From eqn.(1.47), we can conclude that $E_{F1} = E_{F2}$. In other words, there is no discontinuity in the Fermi level at thermal equilibrium, or more generally, **the equilibrium Fermi level is constant throughout the materials in intimate contact**. Another way of expressing this is to say that there is no gradient in the Fermi level at equilibrium, i.e.

$$\frac{dE_F}{dx} = 0 \ldots\ldots.(1.48)$$

Eqn.(1.48) is an important relation which will be used very often in subsequent chapters.

1.3.10 Concept of Vacuum Level, Work Function and Electron Affinity

Let us consider Einstein's experiments on the photoelectric effect, where he observed that electrons can be emitted from a metal by shining light on it. However, for electron emission, the incident photons needed to have a minimum energy $q\phi_m$ (expressed in eV), called the **work function** of the metal. Let us try to explain this phenomenon with the help of energy band diagram. In a metal, considering that the conduction band and valence bands overlap, we can assume that all the states below Fermi level are occupied by electrons, while all the states above E_F are empty. Therefore when an electron at E_F absorbs an energy equal to $q\phi_m$, it is raised to an energy level E_{VAC}, called the **vacuum level** as shown in Fig.1.16(a). This electron is now 'emitted' and is therefore free from the forces which had bound it to the metal. It is also clear from this figure, that $q\phi_m = E_{VAC} - E_F$ is the minimum energy required to emit an electron

since all the states above E_F are empty. The work function is therefore a measurable quantity for the metal. For example, the value of $q\phi_m$ is 4.1eV for Al, while for Au it is 5.0eV.

In the case of semiconductors, however, E_F lies in the forbidden gap, unless the semiconductor is degenerately doped. Therefore, there are no electrons at E_F. The electrons which are emitted are usually from the bottom of the conduction band. The energy difference between the Vacuum Level (E_{VAC}) and the conduction band edge (E_C) is called the **electron affinity** ($q\chi_s$ in eV) of the semiconductor and is also the minimum energy required to emit an electron from the semiconductor as shown in Fig.1.16(b). This is again a measurable quantity and is a property of a particular semiconductor. For example, the value of $q\chi_s$ is 4.15eV for Si. On the other hand, the work function of a semiconductor is not fixed but depends on the position on the Fermi level, which in turn depends on the doping concentration.

The concept of Vacuum Level is very useful for drawing the energy band diagram of multi-material systems, where E_{VAC} is taken as the reference for drawing the energy band diagram of the individual materials. This will be clear when we discuss heterojunctions and MOSFETs in later chapters.

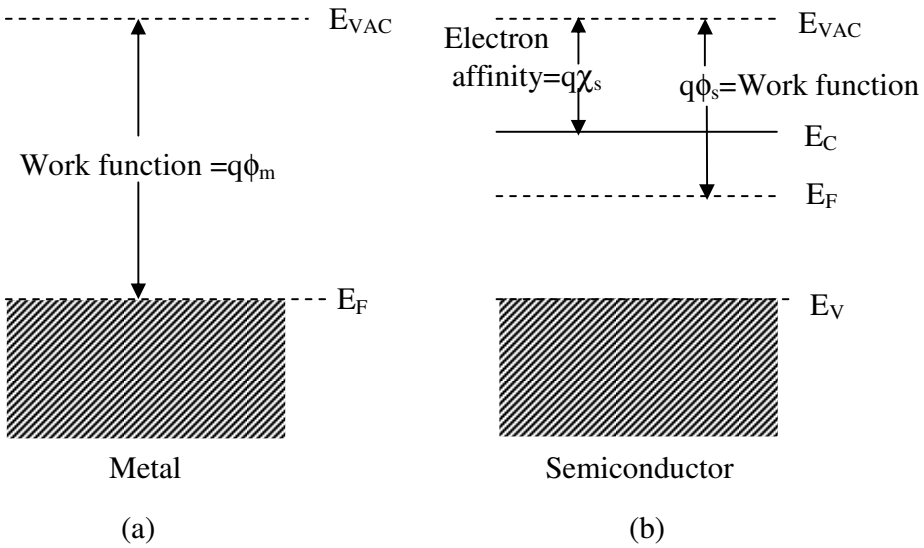


Fig. 1.16 Band diagram of (a) metal and (b) semiconductor showing work function and electron affinity